



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Understand Percent

Lesson 4-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

Language Objective: I can explain what a percent means using the words per hundred, part, whole, and equivalent ratio.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent

Per hundred

Decimal grid

Part-to-whole

Greater than 100%



Tap any word to see its meaning and a picture.

1 The word that means “per hundred” is .

2 The meaning hidden inside the word percent is
.

3 A 10-by-10 grid of 100 squares used to show a percent is a
 .

4 A comparison of one part against the whole amount is a
 comparison.

5 A percent above 100 means the amount is than one whole.

Write About the Math

What percent of the gray kitten's weight does the orange kitten weigh, and how does the tape diagram show it?

¿Qué porcentaje del peso del gatito gris pesa el gatito naranja, y cómo lo muestra el diagrama de cinta?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Percent**
Porcentaje

☐ **Per hundred**
Por cada cien

☐ **Decimal grid**
Cuadrícula decimal

☐ **Part-to-whole**
Parte a todo

☐ **Greater than 100%**
Mayor que 100 %

Model: 150% is more than one whole, and 0.5% is a tiny piece smaller than 1%. — Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The gray kitten is the ____, so its weight is ____%. The orange kitten needs one whole tape plus ____ of another. Adding the parts, ____% + 50% = ____%. A percent greater than 100% means the orange kitten weighs ____ than one whole gray kitten.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- My evidence is ____, so ____.

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Per hundred
3. **Notes 3:** Decimal grid
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Greater than 100%
6. **Watch for this mistake:** *A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor — $10 \times 10 = 100$, so $7 \times 10 = 70$, giving 70%. Reading it as 7% describes seats that are nearly empty when they are nearly full. A second mistake is treating a percent as an amount: 150% of a 200-point goal is $1.5 \times 200 = 300$ points, not 150 points.*
7. **Try It 1:** D. Twice as much as you need — 100% is what you need, so 200% is two of those — twice as much.
8. **Try It 2:** — Dividing a percent by 100 tells you how many wholes it is.